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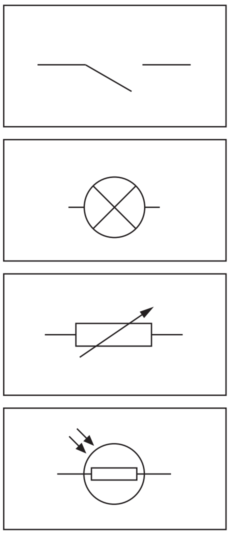
Answer **all** questions.

1. Students are investigating the properties of series and parallel circuits.

(a) **Draw a straight line** linking the circuit symbol on the left with the name of the component on the right. [3]



lamp

switch

variable resistor

LDR

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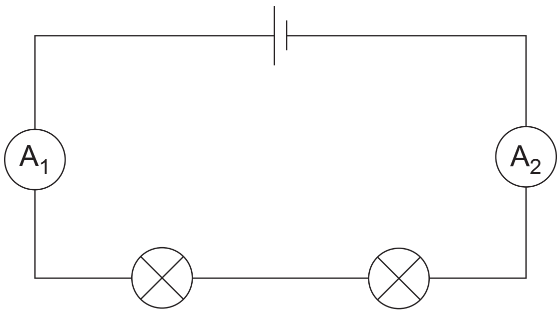
(b) The students set up the following circuit and measure the current in different places.



(i) <u>Underline</u> words or phrases from the brackets to correctly complete the following sentences. [2]

The circuit is a (**double** / **series** / **parallel**) circuit.

The current reading on ammeter A_2 is (**less than** / **the same as** / **more than**) the reading on ammeter A_1.

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(ii) The battery voltage in the circuit is 3 V and the current through the lamps is 0.25 A.

I. Use the equation:

power = voltage × current

to calculate the power of the circuit. [2]

power = ... W

II. Use the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the circuit. [2]

resistance = ...

III. **Circle** the correct unit for resistance. [1]

J Ω kg

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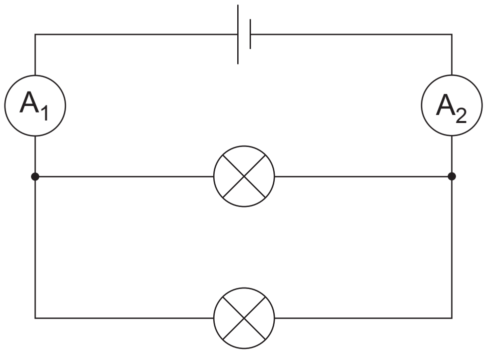
(c) The students now use the **same** components to make the following circuit.



The students notice that the lamps are **brighter**.

Select words or phrases from the box to correctly complete the sentences below.
You may use each word once, more than once or not at all. [2]

increases	decreases	stays the same

The lamps are brighter because the resistance of the circuit ....

so the current through each lamp ....

3420U101
05

12